refinance (against the G-Secs.).

PRISONER'S DILEMMA

A popular example in *game theory* which concludes why co-operation is difficult to achieve even if it is mutually beneficial, ultimately making things worse for the parties involved. It is shown giving an example of two prisoners arrested for the same offence held in different cells. Each prisoner has two options, i.e., confess, or say nothing. In this situation there are *three* possible outcomes:

(i) One could confess and agree to testify against the other as a state witness, receiving a light sentence while his fellow prisoner receives a heavy sentence.

(ii) They can both say nothing and may turn out to be lucky getting light sentences or even be let off due to lack of firm evidence.

(iii) They may both confess and get lighter individual sentences than one would have received had he said nothing and the other had testified against him.

The second outcome looks the best for both the prisoners. However, the risk that the other might confess and turn state witness is likely to encourage both to confess, landing both with sentences that they might have avoided had they been able to co-operate by remaining silent.

In reality, firms behave like these prisoners, not setting prices as high as they could do if they only trusted the other firms not to undercut them. Ultimately, the firms are worse off i.e. all firms suffer.

POPULATION TRAP

A situation of population growth rate greater than the achievable economic growth rate. This makes it difficult to alleviate poverty;–government is suggested to implement population control measures.